

Appendix A Lab-Board I/O Devices

Device	Port	Driver
Buzzer	PE2	PIO
Brushless motor	PB5	OC1C/OC0A
Up, Right, Down, Enter, Left	PA3-7	PIO
Ethernet UART	PD2-3	UART receiver, transmitter
External connector	PA0-7 PC0-7	
External interface direction	PD5-6	PIO
Fan speed	PD7	Timer0 clock input
Frequency in	PD4	ICP1
Heater	PB7	OC
Humidity sensor	PF1	ADC1
Keypad (3 x 4)	PA0-6	PIO
Lamp	PB6	OC1B
LCD display	PD0-1	I2C SCL, SDA (address 0x28)
LEDS (8)	PC0-7	PIO
LED RGB	PD0-1	I2C SCL, SDA (address 0x66)
Light sensor	PF0	ADC0
Memory	PD0-1	I2C SCL, SDA (address 0x57)
Microphone	PF2	ADC2
Motion sensor (PIR)	PE5	INT5
PORTA input select	PE0-1	PIO (See table for selection)
Push buttons (3)	PA0-2	PIO
Real time clock	PB0-3	SPI SS, SCK, MOSI, MISO
Real time clock interrupt	PE4	INT4
Scroll wheel	PE6-7	INT6-7
Seven segment display (2)	PC0-7	PIO
Speaker	PB4	OC2A
Toggle switches (8)	PA0-7	PIO
Temperature	PF3	ADC3
USB UART	PD2-3	UART receiver, transmitter
USB micro	PE3	
Variable resistor 1	PF1	ADC2
Variable resistor 2	PF2	ADC1
Microcontroller Systems		